

OMAR ALBARES REYES

Cross-Platform Software Developer · Applied AI

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PROFESSIONAL SUMMARY

Cross-platform software developer with 4+ years of IT experience, working with Python, PHP/Laravel, Rust, and Flutter. Specialized in AI agents (tool calling, RAG, hybrid search, and persistent memory), offline-first architectures, and real-time sync. Focused on system reliability, automated testing, and user experience.

WORK EXPERIENCE

AI & Full-Stack Developer Intern | Enteracloud MX

Sep 2026 – Present

CosechIA project: a vineyard and winery management platform for Baja California with a domain-specific AI agent. *PHP, Laravel, Vue, Inertia, PostgreSQL + pgvector, Tailwind, Pest*

- Built an AI agent (“Vini”) from scratch with a custom tool-calling loop and 7 domain tools (weather, disease risk, grape maturity, recommendations, vineyard log, and knowledge search).
- Implemented RAG with hybrid search: vector (pgvector embeddings) + lexical BM25, merged with Reciprocal Rank Fusion, producing answers that cite their sources.
- Developed versioned working and long-term memory, plus a proactive agent that evaluates rules and generates decisions and notifications.
- Designed a 50+ table data model (vineyards, blocks, phenological cycles, lots, wineries, logs) with role-based dashboards and weather integration (Open-Meteo).
- Enforced per-grower data isolation with policies, backed by 50+ Pest test suites and static analysis (Larastan).

Service Desk Analyst | Antlia Systems

Jan 2023 – Present

- Manage technical incidents and maintain the internal knowledge base.
- Provide ongoing user support and improve internal processes; built Autum so no ticket goes unattended.

FEATURED PROJECTS

M.E.I.D.O + Meido Mobile – Autonomous AI assistant ecosystem | Python, PyQt6, aiohttp, Flutter, Convex, Whisper

- Desktop agent for Arch Linux + Hyprrland with ~150 function-calling tools, per-request dynamic tool selection, and human approval for system actions.
- On-device GPU voice: Faster-Whisper speech recognition (~0.15 s per phrase) and streaming XTTS-v2 synthesis with a cloned voice.
- Flutter Android app that runs without the PC and syncs chat, vector memory, and reminders in real time via Convex.

Mitpad – Offline-first note-taking app | Rust, Tauri, SvelteKit, TypeScript, SQLite, Convex

- Desktop app that works offline first, with a local sync queue and idempotent mutations to the cloud.
- Migrated sync from Neon to Convex and added interactive graphs of note relationships.

Autum – Ticket monitor for ServiceDesk Plus | Node.js, Playwright, Telegram Bot API

- Detects unassigned tickets every minute and sends Telegram alerts with persistent reminders until acknowledged; in daily use at a help desk.

Lab Management System | Python, Django, PostgreSQL, Flutter, Render

- Django API and web dashboard with a Flutter mobile app; refactored the codebase, optimized database queries, and redesigned the web UI.

TECHNICAL SKILLS

- **Languages:** Python, PHP, Rust, JavaScript, TypeScript, Dart, Java, C#, SQL, HTML, CSS
- **Frameworks:** Laravel, Django, Vue, Inertia, Tauri, SvelteKit, Flutter, PyQt6, aiohttp, Tailwind
- **AI:** Tool-calling agents, RAG, embeddings, hybrid search (pgvector + BM25), LLMs (OpenRouter, Ollama), Whisper, TTS
- **Databases & Cloud:** PostgreSQL, pgvector, SQLite, Convex, Supabase, Neon, Render
- **Tools & Practices:** Git/GitHub, REST/HTTP APIs, offline-first, testing (Pest, pytest), Playwright, Linux (Arch)

EDUCATION & LANGUAGES

- **Associate Degree (TSU) in Cross-Platform Software Development** | Universidad Tecnológica de Tijuana (UTT) | Sep 2024 – Dec 2026 (expected)
- **Languages:** English (C1, Advanced), Spanish (Native)